

1 Purpose and objective

Develop an RFID-enabled library solution that integrates with an existing Integrated Library Management Software system without changing, deleting, or corrupting existing records. The software must support library operations, RFID workflows, reporting, data migration, testing, documentation, offline maintenance, and user training.

The candidate team must translate these requirements into a modular prototype and a production-oriented implementation plan. Where an external device, vendor SDK, existing database schema, or live service is unavailable, the team must define an adapter interface and supply a working mock.

2 Expected outcome

- A web-based library application or demonstrable extension covering the required functional workflows.
- An RFID integration or middleware service with mock adapters for staff readers, handheld readers, security gates, smart cards, cameras, email, SMS, and print services.
- A controlled migration utility for approximately 20,000 source records supplied in spreadsheet format.
- A requirements traceability matrix, architecture package, test evidence, deployment guide, administration guide, user guide, and demonstration.
- A solution that can be activated, updated, upgraded, and supported in an offline or isolated environment.

3 Scope boundaries

Included software work	Excluded from candidate implementation
Library workflows, web interface, database, search, reports, RFID middleware, migration, security controls, test automation, documentation, and demo.	Manufacture, physical installation, or repair of RFID readers, gates, tags, smart cards, printers, PCs, servers, cameras, or cabling.
Mocked integration with RFID devices and the existing ILMS through documented interfaces.	Access to production systems, production databases, personal data, restricted networks, or any live institutional environment.
Offline deployment design and update package.	Claims of NCIP, SIP2, ISO, OEM, security, or performance certification without test evidence or authorised certification.

4 Functional requirements

ID	Area	Requirement
FR 01	Core ILMS	Support acquisition, cataloguing, serials, circulation, OPAC, barcode and spine-label workflows, and report generation.
FR 02	Web and search	Provide a web interface, full-text search, search engine, real-time indexing, net cataloguing, and virtual bookshelf.

ID	Area	Requirement
FR 03	RFID interoperability	Integrate staff station, handheld reader, security gate, smart card, and item tags through middleware. Support NCIP 2.0 and an SIP2 adapter boundary.
FR 04	Circulation	Support check-out, check-in, renewal, due-date enquiry, patron-card personalisation, reference-book restrictions, member blocking, configurable fine limits, and role-based circulation rights.
FR 05	Tagging	Validate title or member records before tagging or re-tagging. Associate each RFID tag with its bibliographic or item record and support tag monitoring.
FR 06	Inventory	Support stock verification, shelf management, misplaced or missing item identification, bulk reading, and audible or visible confirmation for handheld operations.
FR 07	Security gate events	Detect unauthorised or unissued items, read the security bit while the LMS is offline, record accession numbers, and integrate alarm, footfall, CCTV photo, and email-notification events.
FR 08	Dashboard and reports	Provide library-usage statistics and reports for tagged items, members, circulation, operator or RFID client activity, fines, and gate events.
FR 09	Notifications	Provide configurable email, SMS, and print notifications per user through provider-neutral adapters.
FR 10	Data migration	Import approximately 20,000 records from spreadsheet files with validation, duplicate handling, error logs, reconciliation, and rollback.
FR 11	Administration	Provide user and role management, configuration, audit logs, software-version reporting, and system health information.
FR 12	Offline lifecycle	Support offline activation, updates, upgrades, patches, and security updates. Produce an installable update package with rollback instructions.

5 Nonfunctional and operational requirements

ID	Area	Requirement
NFR 01	Data integrity	Back up the existing database before integration. The solution must not modify, delete, or corrupt existing library records.
NFR 02	Security and privacy	Protect member, circulation, and institutional data. Use least privilege, secure secrets, audit logging, input validation, and encrypted transport where applicable.
NFR 03	Compatibility	Target Windows 11 clients, Windows Server 2022 or later, Ethernet networking, and a deployment model suitable for an isolated site.
NFR 04	Licensing	Identify all dependencies and licences. Pin dependency versions and document redistribution or offline-use restrictions.
NFR 05	Testability	Provide functional, integration, performance, migration, security, regression, and user-acceptance test evidence.
NFR 06	Maintainability	Use modular adapters, configuration outside code, structured logs, versioned database migrations, documented builds, and rollback procedures.
NFR 07	Supportability	Design diagnostics, health checks, structured logs, and runbooks for software and integration incidents.
NFR 08	Documentation	Provide system architecture, device configuration, IP and interface details, API documentation, software versions, installation steps, administration procedures, user procedures, and troubleshooting guidance.
NFR 09	Delivery	Complete the candidate exercise from D0 to D10 and identify all work remaining for production readiness.

6 Required solution architecture

The submission must use a layered design. Candidates may select the programming languages and frameworks, but must explain their choices and demonstrate the following boundaries.

Layer	Minimum responsibility
User layer	Staff web interface, OPAC or search interface, dashboard, reports, and administration.
Application layer	Cataloguing, circulation, member, fines, inventory, notification, reporting, and access-control services.
RFID middleware	Device-neutral commands and events for staff reader, handheld reader, gate, smart card, printer, and camera.
Interoperability	NCIP 2.0 adapter, SIP2 adapter boundary, and documented mock interfaces where vendor endpoints are unavailable.
Data layer	Transactional database, search index, audit log, migration staging area, and backup or restore process.
Integration layer	Email, SMS, print, CCTV-photo, and external ILMs connectors.
Operations layer	Configuration, secrets, offline installer, logs, health checks, updates, backups, and rollback.

7 Mandatory deliverables

ID	Deliverable	Contents
D1	Requirements package	Assumptions, clarification log, use cases, data dictionary, and requirement-to-feature traceability matrix.
D2	Architecture package	Context, component, deployment, database, interface, and security designs.
D3	Working prototype	Runnable application and mock integrations demonstrating the priority acceptance scenarios.
D4	Source repository	Source code, commit history, branch strategy, dependency lock files, sample configuration, and licence inventory.
D5	Database and migration	Schema, versioned migrations, sample synthetic dataset, import utility, validation report, reconciliation report, and rollback.
D6	Test package	Test plan, automated tests, manual scenarios, defect log, performance results, security checks, and UAT evidence.

ID	Deliverable	Contents
D7	Deployment package	Offline installer or container package, environment checklist, configuration guide, backup, restore, update, and rollback procedures.
D8	Operational documentation	Administrator guide, user guide, API documentation, troubleshooting runbook, and training plan.
D9	Demonstration	Recorded or live demonstration, presentation, known limitations, and production-completion backlog.

8 Priority acceptance scenarios

ID	Acceptance scenario
AC 01	Import a supplied spreadsheet into staging, reject invalid rows, identify duplicates, migrate valid records, and reconcile source and target counts.
AC 02	Create or locate a bibliographic item, validate it, associate a mock RFID tag, and display the tag-to-item relationship.
AC 03	Check out, renew, and check in an item through a mocked NCIP or SIP2 flow while preserving an auditable transaction history.
AC 04	Block circulation for a reference item, a blocked member, or a member whose fine exceeds a configurable limit.
AC 05	Use a handheld-reader mock to inventory a shelf and report missing and misplaced items with visible and audible confirmation events.
AC 06	Generate an unauthorised-removal event from the gate mock, record the accession number, attach a mock CCTV image, and queue an email notification.
AC 07	Demonstrate staff smart-card login through a mock credential provider and enforce role-based permissions.
AC 08	Generate dashboard statistics and filterable reports for tagged items, members, circulation, operators, RFID clients, and gate events.
AC 09	Install and run the solution without internet access, apply an offline update, and complete a documented rollback.
AC 10	Restore from backup after a simulated failed migration or integration change and prove that the original data remains intact.

9 Standard operating procedure

Follow the stages below in order. Each stage ends with a reviewable output. Do not begin production-style integration until the preceding review is complete.

Stage 1 Initiation and controls D0

- Create the repository and issue tracker.
- Read this assignment and build the first requirements traceability matrix.
- Record assumptions, questions, risks, dependencies, and excluded work.
- Confirm that all data and credentials are synthetic.

Stage 2 Requirements and design D1 to D2

- Prepare use cases and acceptance criteria.
- Define the domain model, database schema, APIs, event model, integration contracts, security controls, and deployment topology.
- Create mock contracts for unavailable ILMS and RFID interfaces.
- Hold a design review and record decisions.

Stage 3 Foundation build D3

- Implement project structure, configuration, authentication, role management, audit logging, health checks, database migrations, and test framework.
- Prepare synthetic seed data and a repeatable local build.
- Run static checks and baseline tests.

Stage 4 Core library functions D4 to D5

- Implement cataloguing, members, circulation, fines, restrictions, search, reporting, and dashboards.
- Demonstrate key user journeys and record defects.
- Update the traceability matrix and API documentation.

Stage 5 RFID middleware and integrations D6

- Implement device-neutral commands and event handlers.
- Add NCIP 2.0 and SIP2 adapter boundaries with mocks.
- Implement tagging, inventory, gate-event, notification, smart-card, and camera workflows.
- Test retries, timeouts, duplicate events, and offline behaviour.

Stage 6 Migration and resilience D7

- Profile the sample spreadsheet.
- Run validation and dry-run import.
- Reconcile counts and errors.
- Execute migration into the test database, then demonstrate rollback and restore.

Stage 7 Verification D8

- Run functional, integration, regression, performance, security, migration, backup, restore, offline-update, and rollback tests.
- Map every test result to a requirement.
- Triage defects by severity and retest fixes.

10 Engineering controls

Control	Required practice
Version control	Use feature branches and reviewed merge requests. Keep meaningful commits. Tag the final evaluation release.
Code quality	Apply formatter, linter, static analysis, dependency scanning, and automated tests in the build pipeline.
Configuration	Keep environment-specific values outside source code. Supply a sample configuration without secrets.
Secrets	Do not commit passwords, tokens, certificates, private keys, or real personal data.
Data protection	Use synthetic data. Mask logs. Apply least privilege. Record every administrative and circulation action.
Dependencies	Pin versions and provide a software bill of materials with licence and offline-installation notes.
Defects	Record severity, reproduction steps, owner, status, fix version, and retest evidence.
Change control	Update the requirement, design, code, test, and documentation together when behaviour changes.